

Programming in MATLAB

Matlab is a commercial "Matrix Laboratory" package which operates as an interactive programming environment

MATLAB is also a programming language. By creating a file with the extension **.m** you can easily write and run programs. If you were to create a program file **myfile.m** in the MATLAB language, then you can make the command myfile from MATLAB and it will run like any other MATLAB function. You do not need to compile the program since MATLAB is an interpretative (not compiled) language. Such a file is called an m-file.

Q#01:

```
>> N = 5; k = -N:N;
>> x = sin(k*pi/N);
>> y = cos(k*pi/N); % x and y - vertices of the polygon to be filled
>> fill(x,y,'g')
>> axis square
>> text(-0.45,0,'I am a green polygon')
```

Q#2:

```
>> figure % bar plot of a bell shaped curve
>> x = -2.9:0.2:2.9;
>> bar(x,exp(-x.*x));
>> figure % staircase plot of a sine wave
>> x = 0:0.25:10;
>> stairs(x,sin(x));
>> figure % errorbar plot
>> x = -2:0.1:2;
>> y = erf(x); % error function; check help if you are interested
>> e = rand(size(x)) / 10;
>> errorbar (x,y,e);
>> figure
>> r = rand(5,3);
>> subplot(1,2,1); bar(r,'grouped') % bar plot
>> subplot(1,2,2); bar(r,'stacked')
>> figure
>> x = randn(200,1); % normally distributed random numbers
>> hist(x,15) % histogram with 15 bins
```

Q#3:

3D line plots

```
>> t = linspace(-10*pi,10*pi,200);
>> plot3(t.*sin(t), t.*cos(t), t, 'md-'); % plot the curve in magenta
>> title('Curve r(t) = [t sin(t), t cos(t), t]');
>> xlabel('x-axis'); ylabel('y-axis'); zlabel('z-axis');
>> grid
```

Q#4:

```
>> [X,Y,Z] = peaks; % create values for plotting the function
>> surf(X,Y,Z); % plot the surface
>> figure
>> contour (X,Y,Z,30); % draw the contour lines in 2D
>> colorbar % adds a bar with colors corresponding to the z-axis
>> title('2D-contour of PEAKS');
>> figure
>> contour3(X,Y,Z,30); % draw the contour lines in 3D
>> title('3D-contour of PEAKS');
>> pcolor(X,Y,Z); % z-values are mapped to the colors and presented as
% a 'checkboard' plot; similar to contour
```

Q#5:

```
>> findstr(t, 'c') % finds a character 'c' in string t; positions are returned
ans =
11 16
>> findstr(t, 'racter') % finds the beginning of a string 'racter' in t
ans =
14
>> findstr(t,u) % finds string u in string t; returns the position
ans =
11
>> strcat(u,u1) % concatenates multiple strings: u and u1
ans =
characterretcarahc
>> strcmp(u,u1) % comparison of two strings;
ans = % 1 for identical strings; 0 for different strings
0
>> q = num2str(34.35) % converts a number into a string
q =
34.35
>> z = str2num('7.6') % converts a string into a number
z =
7.6
>> whos q z % q is now a string, i.e. a character array; z is a number
Name Size Bytes Class
q 1x5 10 char array
z 1x1 8 double array
>> t = str2num('1 -7 2') % converts a string into a vector of number
```

Q#6:

```
t = 0:pi/20:2*pi;
y = exp(sin(t));
plotyy(t,y,t,y,'plot','stem')
xlabel('X Axis')
ylabel('Plot Y Axis')
title('Two Y Axes')
```

Q#7:

```
x = -1:1:1; % Define the range of x
y = x.^3; % Raise each element in x to the third power
%Now that you have generated some data, you can plot it using the MATLAB plotting
tools. To start the %plotting tools, type
plottools (After opening plot window you click double on variable x and y then see)
```

Q#8:

```
x = linspace(-pi,pi,25); % 25 points between -π and π
y = sin(x);
%Using the plotting tools, create a graph of  $y = \sin(x)$ :
Plottools
```

Q#09:

```
x = -10:0.005:40;
y = [1.5*cos(x)+4*exp(-.01*x).*cos(x)+exp(.07*x).*sin(3*x)];
plot(x,y)
```

Q#10:

```
x = 0:pi/100:2*pi;
y = sin(x);
y2 = sin(x-.25);
y3 = sin(x-.5);
plot(x,y,x,y2,x,y3)
The legend command provides an easy way to identify the individual plots:
legend('sin(x)', 'sin(x-.25)', 'sin(x-.5)')
```

Q#11

```
xlabel('x = 0:2\pi')
ylabel('Sine of x')
title('Plot of the Sine Function','FontSize',12)
```

Q#12:

```
x = 0:pi/100:2*pi;
y = sin(x);
y2 = sin(x-.25);
y3 = sin(x-.5);
plot(x,y,x,y2,x,y3)
The legend command provides an easy way to identify the individual plots:
legend('sin(x)', 'sin(x-.25)', 'sin(x-.5)')
```

Q#13 :

```
>> plot(t,sin(t),'b')
>> hold on
>> plot(t,cos(t),'r')
```

Q#14 :

```
>> t = (0:0.01:1)';
>> plot(t,[t t.^2 t.^3])
```

Q#15:

```

A=4;
t=0:.001:1;
w=2*pi*5;
p=A*sin(w*t);
q=A*cos(w*t);
subplot(211)
a=plot(t,p)
subplot(212)
b=plot(t,q)

```

Q#16:

```

x=[5 5 5 5 5 5 5 5 5];
y=x.^2;
n=0:9;
stem(n,y)

```

Q#17:

```

function [x,y]=ball(a,b)
x=a*b;
y=a+b;

```

Q#18:

```

A=4;
t=0:.01:1;
w=2*pi*2;
sine=A*sin(w*t);
stem(t,sine)

```

Q#19:

```

t = - 10:.01:10;
t0 = -3;
y = stepfun(t,t0);% step function means 1
plot(t,y)

```

Q#20:

```

y1=ones(1,50); %0-one row ten column
t1=length(y1);
t=0:t1-1;
stem(t,y1)%discrete time signal,plot is used for continous time signal

```

Q#21:

```

A =1;
w0 =2*pi*2;
phi =0;
a =6;
t =0:.001:1;
x =A*sin(w0*t+phi);
y = exp(-a*t);
z = x.*y;
expsin = [x;y;z];

```

```
plot(t,z)
```

Q#22:

```
A=1;  
w0=2*pi*2;  
phi=pi/2;  
a=6;  
t=0:.01:1;  
sine=A*sin(w0*t+phi);  
expo=exp(-a*t);  
expsin=A*sin(w0*t+phi).*exp(-a*t); %multiplication  
subplot(311)  
plot(t,sine)  
subplot(312)  
plot(t,expo)  
subplot(313)  
plot(t,expsin)
```

Q#23:

```
A=1;  
w0=2*pi*2;  
phi=0;  
a=6;  
t=0:.001:1;  
sine=A*sin(w0*t+phi);  
expo=exp(-a*t);  
expsin=A*sin(w0*t+phi).*exp(-a*t);  
figure(1)  
plot(t,sine)  
figure(2)  
plot(t,expo)  
figure(3)  
plot(t,expsin)
```

Q#24:

```
function result= test(n)  
sum=0;  
for i=1:1:n  
sum = sum +i;  
end  
sum
```

Q#25:

```
function result= test(n)  
sum=0;  
for i=1:2:n  
sum = sum +i;  
end  
sum
```

Q#26: [Conditional Expressions]

```
x=10;  
y=20;
```

```

disp(x < y);                % displays 1
disp(x <= 10);              % displays 1
disp(x == y);               % displays 0
disp((0 < x) & (y < 30));   % displays 1
disp((x > 10) | (y > 100)); % displays 0

```

INPUT THE VALUE FROM COMMAND WINDOW

Q#27: [If Statement]

```

if (x < 10)
    disp(x);                % only displays x when x < 10
end
if ((0 < x) & (x < 100))
    disp('OK');             % displays OK if x between 0 and 100
end

```

Q#28: [If Statement]

```

if (a < b)
    if (a < c)
        disp(a);
    else
        disp(c);
    end
else
    if (b < c)
        disp(b);
    else
        disp(c);
    end
end

```

Q#29: [If Statement]

```

if (n <= 0)
    disp('n is negative or zero');
elseif (rem(n,2)==0)
    disp('n is even');
else
    disp('n is odd');
end

```

Q#30: [While Statement]

```

while (p < n)
    p = 2 * p;
end
disp(p);

```

Q#31: [While Statement]

```

i=1;
while (i<=10)

```

```
    disp(i);  
    i = i + 1;  
end
```

Q#32: [For Statement]

```
for i=1:10  
    disp(i);  
end
```

Q#33:

Write a MATLAB code with For statement calling a function from the command window to present the approximate original signal for the following Fourier series:

$$f(t) = 2.5 + \frac{10}{\pi} \left(\sin \frac{\pi t}{4} + \frac{1}{3} \sin \frac{3\pi t}{4} + \frac{1}{5} \sin \frac{5\pi t}{4} + \frac{1}{7} \sin \frac{7\pi t}{4} + \dots \right)$$